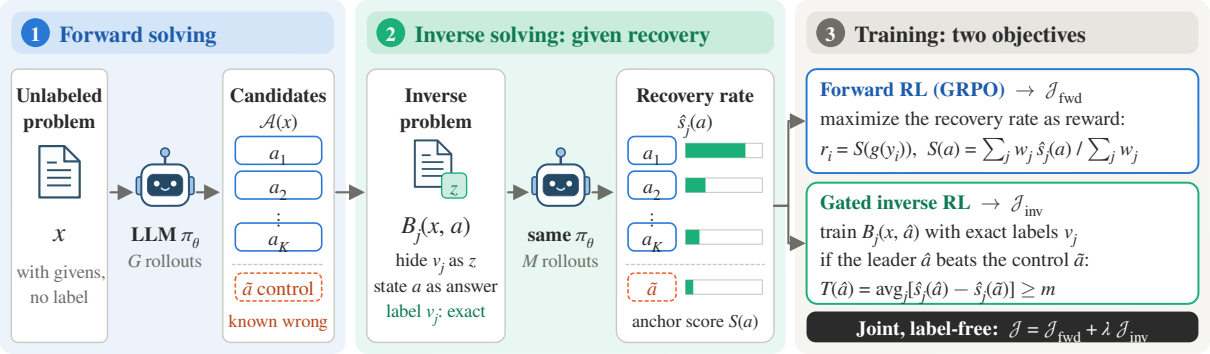
1 Forward solving
Unlabeled problem
x
with givens, no label
LLM π_θ
G rollouts
Candidates
$\mathcal{A}(x)$
a_1
a_2
a_K
$\tilde{a}$ control
known wrong
2 Inverse solving: given recovery
Inverse problem
z
$B_j(x, a)$
hide v_j as z
state a as answer
label v_j: exact
same π_θ
M rollouts
Recovery rate
$\hat{s}_j(a)$
a_1
a_2
a_K
$\tilde{a}$
anchor score $S(a)$
3 Training: two objectives
Forward RL (GRPO) $\rightarrow$ $\mathcal{J}_{\text{fwd}}$
maximize the recovery rate as reward:
$r_i = S(g(y_i))$, $S(a) = \sum_j w_j \hat{s}_j(a) / \sum_j w_j$
Gated inverse RL $\rightarrow$ $\mathcal{J}_{\text{inv}}$
train $B_j(x, \hat{a})$ with exact labels v_j
if the leader $\hat{a}$ beats the control $\tilde{a}$:
$T(\hat{a}) = \text{avg}_j[\hat{s}_j(\hat{a}) - \hat{s}_j(\tilde{a})] \geq m$
Joint, label-free: $\mathcal{J} = \mathcal{J}_{\text{fwd}} + \lambda\, \mathcal{J}_{\text{inv}}$